

Bridging the Trust Gap in Financial Fraud Detection: An Explainable AI Framework for Regulatory Compliance

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Abstract—Credit card fraud detection systems faced a fundamental conflict between classification accuracy and the regulatory transparency obligations imposed by emerging global financial laws. Existing models optimized purely for predictive performance failed to produce the audit-ready documentation required under high-risk AI regulations, creating significant barriers to legal deployment in financial services. This research proposed and evaluated a two-phase explainable ensemble learning framework that integrated Synthetic Minority Over-sampling, a hierarchical stacking ensemble combining Random Forest and Extreme Gradient Boosting, and dual post-hoc interpretability layers using SHapley Additive exPlanations and Local Interpretable Model-agnostic Explanations. The experimental protocol crossed four model types, three feature selection strategies, and two resampling conditions, producing 24 configurations evaluated on the Kaggle credit card fraud benchmark containing 284,807 transactions. Random Forest achieved the highest F1 score of 0.8586 with precision 0.8817 and ROC-AUC 0.9794, correctly identifying 82 of 98 fraud cases in the 56,962-transaction test set. The Stacking Ensemble achieved the highest ROC-AUC of 0.9838. SHAP identified V14, V12, and V4 as the three most influential fraud predictors; LIME produced a fully interpretable explanation at maximum fraud certainty. The results established that regulatory transparency and competitive classification performance were simultaneously achievable within the same pipeline.

Index Terms—Explainable AI, Credit Card Fraud Detection, SHAP, LIME, Ensemble Learning, Regulatory Compliance, Class Imbalance, EU AI Act

I. INTRODUCTION

Artificial Intelligence integration in financial services created a critical trust problem spanning consumers, regulators, and institutions globally [1], [2]. Despite demonstrable improvements in fraud detection accuracy, only 32 percent of consumers trusted automated systems with sensitive payment information [1], representing a fundamental adoption barrier. Systematic analysis of AI integration trends confirmed that regulatory uncertainty and transparency obligations constituted the primary institutional barrier to broader deployment, with compliance gaps generating significant legal and reputational risk for early adopters [2]. Regional asymmetries in regulatory maturity complicated global deployment, with mature markets

demanding strong accountability guarantees while emerging markets exhibited openness reflecting limited regulatory experience.

Credit card fraud detection operated as a complex classification task across severely imbalanced transaction data, where fraudulent events constituted a fraction of one percent of all records. Comprehensive reviews established that tree-based ensemble methods consistently outperformed alternative classifiers on financial benchmarks [9], [10]. However, this architectural strength produced an equally persistent transparency deficit: models achieving near-perfect discrimination categorically failed mandatory transparency audits because their internal logic remained inaccessible to human reviewers, preventing institutions from justifying individual decisions to regulators, customers, or compliance teams.

SHapley Additive exPlanations [11] and Local Interpretable Model-agnostic Explanations [12] emerged as the primary technical response to this transparency deficit. SHAP distributes predictive credit among all input features using cooperative game theory [11]; LIME constructs locally linear surrogate models through perturbation sampling, producing feature weight vectors for individual predictions [12]. A comprehensive review of XAI interpretability methods confirmed these two approaches as dominant in applied financial AI due to their model-agnostic properties [29]. However, the relationship between generating explanations and formally satisfying regulatory requirements remained unresolved.

Three critical gaps persisted. No standardised metric existed for measuring whether XAI outputs satisfied regulatory requirements or remained comprehensible to non-technical stakeholders [3], [4], [26], [27]. A bibliometric analysis confirmed that the compliance gap was the most consistently cited unmet need across XAI finance research [27], while a benchmarking survey showed explanation fidelity varied substantially with no universally reliable approach [28]. An independent assessment framework confirmed that current practices were insufficient to formally establish whether explanations satisfied legal obligations [5], and a systematic evaluation review found the field lacked the instruments needed to

link explanation quality to compliance standards [30].

These limitations carried severe regulatory and commercial consequences. EU AI Act Articles 10 and 13 imposed mandatory data governance, system documentation, and individualised explanation obligations for every automated high-risk decision [6]. Behavioral research established that perceived transparency was a stronger adoption predictor than actual model accuracy [7], while independent analysis identified unexplainable behavior as the primary commercial integration barrier across financial institutions [8]. Compliance-oriented explainability constituted a categorically different design objective from performance-oriented interpretability [26], requiring dedicated architectural solutions.

This research addressed these vulnerabilities by designing a two-phase framework that embedded regulatory compliance requirements into the pipeline from the outset, combining SMOTE-based imbalance mitigation, hierarchical stacking [9], [10], and dual-layer explainability [11], [12] across 24 configurations on the Kaggle credit card fraud benchmark.

A. Main Contributions

- Proposed a regulatory-aligned stacking ensemble with dual SHAP and LIME layers, producing EU AI Act-compliant transparency records for individual fraud predictions.
- Designed a 24-configuration multidimensional evaluation protocol crossing four model types, three feature selection strategies, and two resampling conditions.
- Developed a two-phase pipeline isolating model selection from explainability generation, eliminating test data leakage.
- Empirically validated attribution-based SHAP feature selection against Pearson correlation across identical model configurations.
- Generated LIME instance-level audit records satisfying EU AI Act Article 13 individualised explanation requirements.

II. LITERATURE REVIEW

Research on explainable AI for financial fraud detection has grown considerably but remains structurally fragmented. Studies in this space can be grouped into three methodological clusters: applied XAI for fraud classification, regulatory compliance and privacy, and class imbalance mitigation with evaluation methodology. What is significant is not what each cluster achieved individually, but rather what the three clusters collectively failed to produce together — a unified, compliance-ready pipeline. This section examines each cluster critically, comparing studies against each other and identifying the specific architectural gaps that motivated the present work.

A. Applied XAI for Fraud Classification: Technical Progress Without Regulatory Grounding

The most directly relevant prior work is Keerthana et al. [13] and Iqbal et al. [14], and a close comparison of the two reveals a structural contradiction at the heart of this cluster. Keerthana et al. demonstrated that explanation fidelity was model-dependent — tree-based architectures produced more stable and consistent SHAP attributions than neural counterparts on the same benchmark dataset. This finding has an important implication: choosing the wrong architecture for accuracy reasons could degrade explanation reliability. Yet Iqbal et al., whose work followed chronologically, selected their ensemble architecture exclusively on predictive grounds, achieving strong baselines of RF F1=0.9159 and XGBoost AUC=0.9997, without assessing whether that selection preserved the explanation stability that Keerthana et al. had shown was architecture-dependent. In other words, the field's best-performing ensemble study ignored a key insight from the field's best explainability study. This gap, optimizing for accuracy while implicitly assuming explanation quality follows, is precisely the architectural flaw that the present study corrected by co-designing the model selection and explainability evaluation phases.

Extending the comparison further, both Iqbal et al. and Keerthana et al. evaluated their outputs against standard classification benchmarks but neither asked the operationally critical question: are these explanations sufficient for a compliance officer to satisfy an EU AI Act audit? Tomy and Ojo [19] attempted to bridge this gap conceptually by framing fraud detection explicitly as an accuracy-interpretability trade-off, but their work remained at the level of framing without producing a deployable pipeline. Vijayanand and Smrithy [15] extended the XAI-ensemble approach to mobile money fraud, which confirmed methodological transferability across transaction types, but the mobile context introduced different regulatory constraints that limited direct comparability. Damanik and Liu [20] pursued the imbalance problem aggressively with K-SMOTEENN combined with stacking ensembles and achieved strong performance gains, yet their complete absence of any explainability component meant that the stronger the model became, the less auditable it was. Taken together, this cluster demonstrates a consistent pattern: each advance in one dimension: accuracy, stability, or imbalance handling; was made without integrating the others. The present study addressed this by treating all three as jointly necessary rather than independently optimisable.

The structural character of this limitation was confirmed quantitatively by the bibliometric analysis of Chen et al. [27], which mapped citation networks across XAI finance research from 2018 to 2024 and found that the compliance gap was the most consistently cited unmet need across the entire field. The taxonomic review by Martins et al. [26] categorised credit fraud detection as the highest-frequency XAI finance use case, yet also identified it as the domain with the widest

gap between technical capability and regulatory deployment. That gap persists not because the tools are unavailable but because no study has designed its architecture around closing it.

B. Regulatory Compliance and Privacy: Framework Without Implementation

The second cluster approached fraud detection from the regulatory rather than the technical direction, and its limitations are the mirror image of the first cluster: where Cluster A had implementation without regulatory grounding, Cluster B had regulatory understanding without implementation. Khan et al. [21] conducted the most comprehensive systematic review mapping model-agnostic XAI methods against international regulatory frameworks, concluding theoretically that local linear approximations could satisfy legislative obligations. However, Khan et al. produced no code, no dataset evaluation, and no evidence that their theoretical mapping held in practice. Theodorakopoulos et al. [22] corroborated the urgency from the risk management side, showing that unexplainable architectures made compliance assessment exceptionally difficult for banking authorities, but again without producing a functioning pipeline.

Aljunaid et al. [17] and Awosika et al. [18] advanced the most technically concrete proposals in this cluster by combining federated learning with SHAP to address both privacy and interpretability simultaneously. Their shared finding, that transparency and privacy are architecturally compatible rather than inherently in tension, was an important theoretical contribution. However, both acknowledged the absence of standardised compliance metrics as the primary unresolved challenge, and neither produced transparency records aligned with specific regulatory articles. The European Banking Authority’s guidelines [6] explicitly required individualised explanation for every automated high-risk decision, a standard that these federated approaches approached in spirit but did not satisfy in documented output format. The result is that the regulatory compliance cluster, despite its awareness of the legislative landscape, remained a collection of frameworks without the implementation machinery to operationalise them.

This gap was further characterised by Calzarossa et al. [5], whose assessment framework established that current measurement practices were insufficient to formally determine whether any given explanation output satisfied legal transparency obligations. Their framework was diagnostic rather than prescriptive, identifying the absence of standardised instruments as a structural problem but not supplying a solution. The present study directly addressed this by producing structured SHAP and LIME outputs compiled into EU AI Act-aligned compliance records, moving from diagnostic awareness to operational documentation.

C. Class Imbalance and Evaluation: The Infrastructure Problem

The third cluster addressed the technical infrastructure underlying both classification performance and explainability reliability. Albalawi and Dardouri [16] provided the most rigorous comparative study in this cluster, evaluating traditional and deep learning methods across multiple imbalance strategies including SMOTE, establishing that SMOTE-based resampling applied exclusively to training data was the most reliable approach for preserving test set integrity. Their finding was directly incorporated into the present study’s preprocessing design. However, Albalawi and Dardouri included no explainability layer, meaning their technically rigorous imbalance study was architecturally incomplete for compliance purposes.

Hafez et al. [23] drew the broadest conclusions from this cluster through a systematic review confirming that class imbalance was the single most consequential methodological challenge in credit fraud detection, and that no reviewed study had successfully combined state-of-the-art imbalance handling with a full compliance documentation pipeline. This finding applied equally to Clusters A and B: every study that handled imbalance well omitted explainability, and every study that addressed explainability treated imbalance as a secondary concern.

The evaluation dimension compounded this problem. A systematic survey of XAI evaluation methods by Nauta et al. [30] established that most published evaluations relied on proxy metrics that did not correspond to any specific regulatory requirement. Weber et al. [24] confirmed through cross-disciplinary analysis that explanation evaluation remained the least developed area across all XAI finance sub-domains. Giudici and Raffinetti [25] provided a partial solution through the SAFE AI methodology, demonstrating that accuracy and interpretability were not fundamentally in conflict, but their framework addressed fairness rather than legal auditability specifically.

The cross-cluster picture is clear. Cluster A could produce explanations but could not ground them in regulatory requirements. Cluster B understood regulatory requirements but could not produce compliant pipeline outputs. Cluster C handled imbalance and evaluated performance but left explainability and compliance unaddressed. The present study was designed specifically at the intersection of all three, combining SMOTE-based imbalance mitigation from Cluster C, stacking ensemble architecture from Cluster A, and explicit EU AI Act Article 10 and 13 alignment from Cluster B, into a single integrated pipeline. Tables I and II summarise the eight most directly relevant studies and a comparative feature analysis.

Table II key: ✓ = implemented; × = absent. [3] = Cerneviene 2024; [5] = Calzarossa 2025; [9] = Ayari 2025; [10] = Nallakaruppan 2024; [21] = Khan 2025; [25] = Giudici 2023. No reference appears in both Tables I and II.

TABLE I
COMPARISON OF RECENT LITERATURE: XAI FOR FINANCIAL FRAUD DETECTION [13]–[20]

Ref.	Approach / Method	Tools / Dataset	Key Contribution	Limitation
[13]	SHAP and LIME on multiple ML architectures for CC fraud	Python, Kaggle CC	First comparative XAI benchmark on fraud; tree models produce most stable attributions	Explanation fidelity measured but not linked to regulatory compliance
[14]	Stacking ensemble with marginal contribution algorithms	RF, XGBoost, Kaggle CC	Published baselines RF F1=0.9159, XGB AUC=0.9997	Architecture selected on accuracy alone; no compliance documentation
[15]	XAI-enhanced ensemble for mobile money fraud	ML classifiers; SAGE mobile	Confirms XAI-ensemble generalisability beyond card transactions	Mobile context only; no compliance pipeline
[16]	Traditional vs deep learning across class imbalance strategies	DL+ML; Kaggle + proprietary	Rigorous multi-model imbalance comparison including SMOTE	No explainability; regulatory requirements absent
[17]	Federated learning with SHAP for privacy-preserving fraud	FL+SHAP; banking data	Privacy and local interpretability are architecturally compatible	High cost; no standardised compliance metrics
[18]	XAI and federated learning for financial fraud	FL+SHAP; financial data	Transparency and privacy co-achievable but no pipeline deployed	No compliance documentation produced
[19]	XAI for CC fraud: bridging accuracy-interpretability gap	ML+XAI; CC dataset	Frames accuracy-interpretability trade-off explicitly	No regulatory mapping; single dataset
[20]	K-SMOTEENN and stacking ensemble for imbalanced fraud	Stacking; Kaggle CC	Strong imbalance handling with ensemble performance gains	No SHAP or LIME; explainability absent

TABLE II
COMPARATIVE FEATURE ANALYSIS OF PROPOSED FRAMEWORK VERSUS RELATED MODELS [3], [5], [9], [10], [21], [25]

Ref.	Class Imbal.	Ensemble	Local XAI	Global XAI	Reg. Compliance	Notes
[3]	×	×	✓	✓	×	
[5]	×	×	✓	✓	✓	
[9]	×	×	×	×	×	
[10]	×	×	✓	✓	×	
[21]	×	×	✓	✓	✓	
[25]	×	×	×	✓	✓	
Proposed	✓	✓	✓	✓	✓	SMOTE + Stacking + SHAP + LIME + EU AI Act docs

III. PROBLEM STATEMENT

Financial institutions deploying automated fraud detection systems faced a fundamental conflict between predictive model accuracy and strict regulatory transparency. Existing fraud detection models, optimized purely for classification performance, failed to produce the audit-ready documentation required by emerging financial regulations. Three main assumptions governed this research: transaction data was available in standard tabular format; historical data patterns reflected real-world European cardholder behaviors; and interpretation tools were applied exclusively post-training. System constraints included the strict unavailability of personal customer details due to PCA anonymization, and the extreme 578:1 class imbalance necessitating synthetic data generation. Critical threats included high false-negative rates allowing actual financial theft to pass undetected, and excessive false-positive rates blocking legitimate customer purchases. Resolving this transparency deficit was necessary to enable safe, legally compliant deployment of automated decision systems in the high-risk

financial sector.

IV. PROPOSED METHODOLOGY

The proposed framework operated in two sequential phases, as depicted in Fig. 1 and Fig. 2. Phase 1 trained and evaluated all 24 experimental configurations to select the best-performing model; Phase 2 applied dual XAI methods exclusively to that best model to generate compliance documentation. This two-phase separation was deliberate: it prevented explanation method choices from influencing model selection, and model selection choices from contaminating explanation evaluation.

A. Dataset Acquisition and Preprocessing

The framework ingested anonymised European credit card records exhibiting extreme class imbalance. Raw financial values possessed vastly different scales that historically destabilised gradient-based optimisation, making preprocessing es-

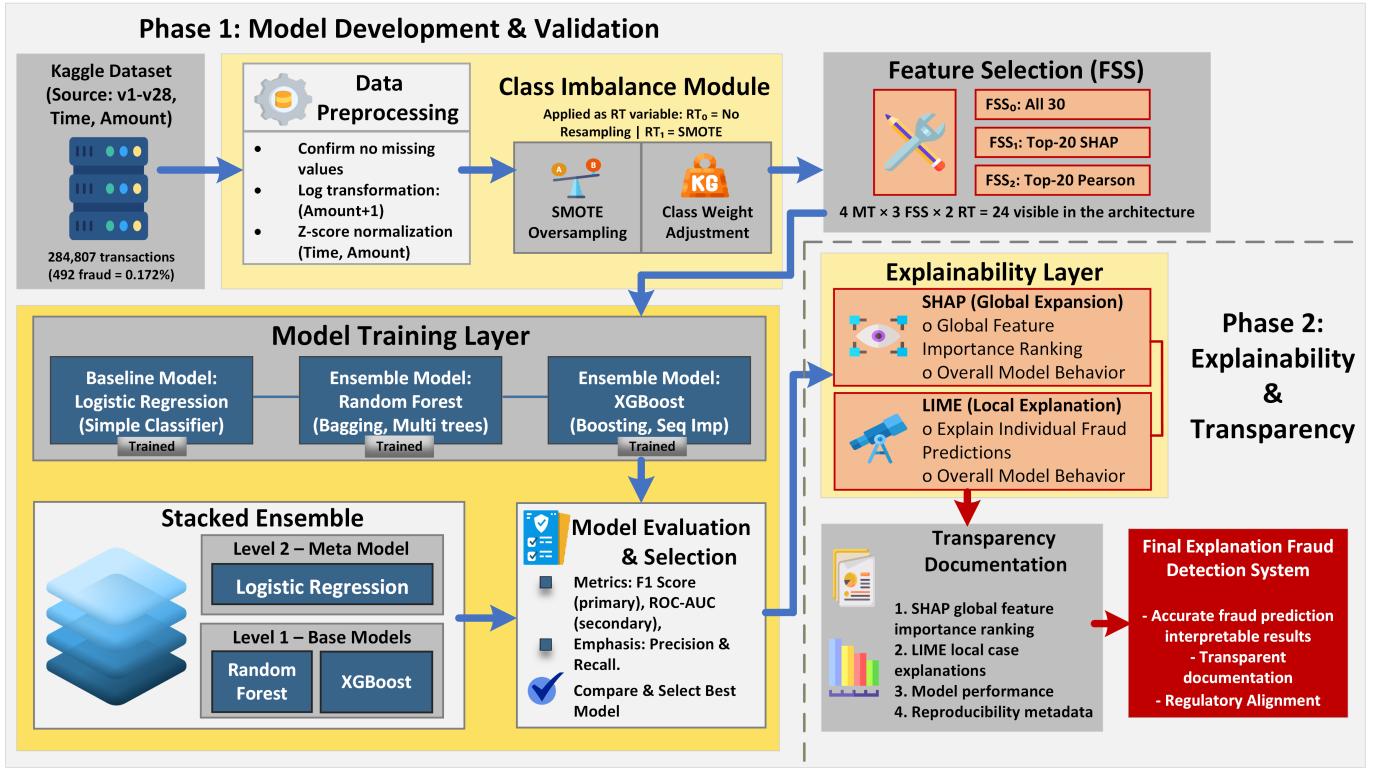


Fig. 1. System architecture of the proposed explainable ensemble framework depicting flow from data processing to compliance generation.

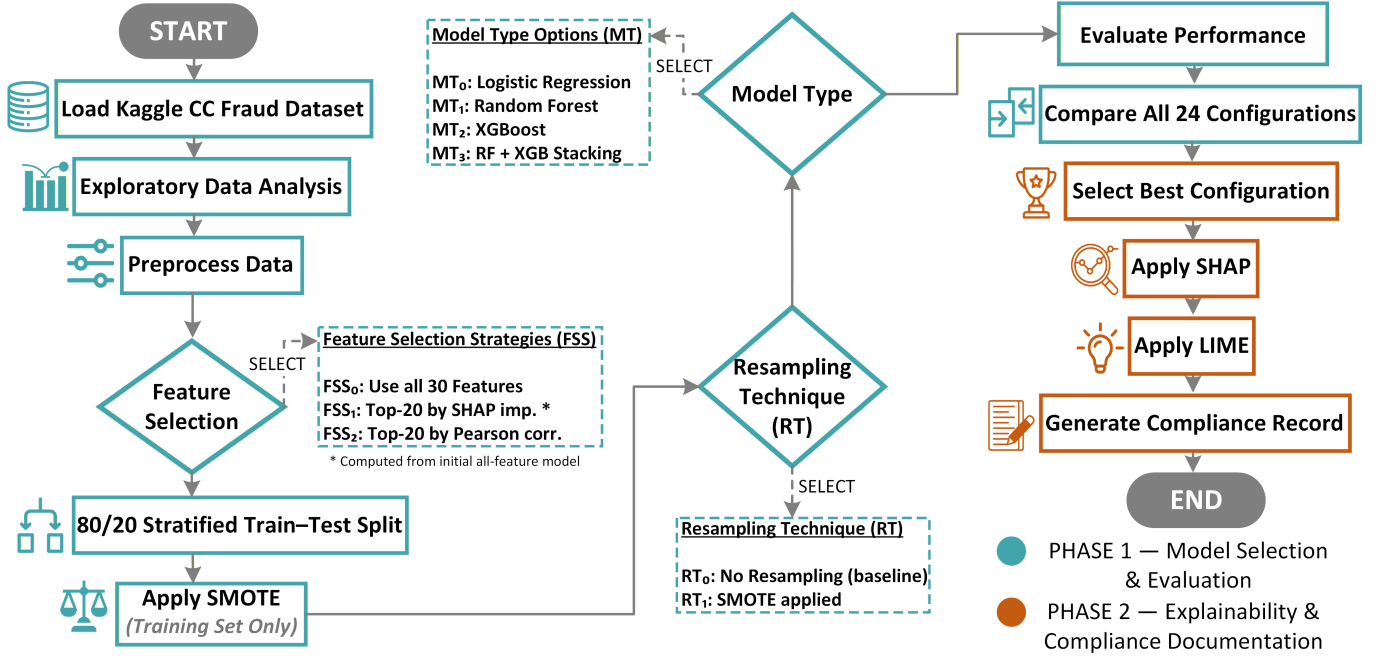


Fig. 2. Operational workflow mapping sequential decision steps to compliance record output.

sential. A logarithmic transformation compressed the right-skewed Amount distribution:

$$x_{\text{amount}} = \log(\text{Amount} + 1) \quad (1)$$

Both Amount and Time were z-score normalised to zero mean and unit variance. These normalised features then flowed into the resampling module, ensuring that all subsequent ensemble models received a numerically stable input space for boundary

construction.

B. Feature Selection and Engineering

Feature selection addressed the noise introduced by 30 features in severely imbalanced data and provided an empirical test of whether attribution-based selection (FSS1) offered measurable advantages over correlation-based selection (FSS2). FSS0 used all 30 original features as the baseline. FSS1 selected the top 20 by global Shapley importance from an initial all-feature model. FSS2 selected the top 20 by Pearson statistical correlation:

$$r = \frac{\sum (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum (x_i - \bar{x})^2 \sum (y_i - \bar{y})^2}} \quad (2)$$

Crossing FSS0, FSS1, and FSS2 against all four model types under both resampling conditions enabled direct, controlled comparison of feature selection strategies within the same experimental grid.

C. Class Imbalance Mitigation

The 578:1 legitimate-to-fraud ratio induced severe majority-class bias in standard classifiers. SMOTE addressed this by generating synthetic fraud instances through k -nearest-neighbour interpolation within the feature space:

$$\tilde{x} = x_i + \lambda \cdot (x_j - x_i), \quad \lambda \sim \mathcal{U}(0, 1) \quad (3)$$

Critically, SMOTE was applied exclusively to the training partition after stratified splitting. This preserved the authentic 0.172 percent fraud rate in the test set across all 24 configurations, ensuring that evaluation metrics reflected genuine deployment conditions rather than an artificially balanced test distribution.

D. Ensemble Model Architecture

Four model types constituted the MT experimental variable, spanning a deliberate complexity gradient from interpretable baseline to full ensemble. MT0 was Logistic Regression as the interpretable linear baseline. MT1 was Random Forest with 100 estimators and balanced class weights. MT2 was Extreme Gradient Boosting with 100 estimators. MT3 was the Stacking Ensemble with RF and XGBoost as Level-1 base learners and Logistic Regression as Level-2 meta-learner, trained via 5-fold cross-validation:

$$\hat{y}_{\text{stack}} = \sigma(w_1 \hat{y}_{\text{RF}} + w_2 \hat{y}_{\text{XGB}} + b) \quad (4)$$

This design captured both independent variance reduction through bagging and sequential residual correction through boosting, with the meta-learner learning the optimal combination of the two complementary strategies.

E. Explainability and Transparency Layer

This terminal Phase 2 component generated audit-ready explanations for the best-performing model. Shapley values [11] distributed each feature's marginal contribution across all possible feature coalitions:

$$\phi_i = \sum_{S \subseteq F \setminus \{i\}} \frac{|S|! (|F| - |S| - 1)!}{|F|!} [f(S \cup \{i\}) - f(S)] \quad (5)$$

These global attributions produced ranked importance scores across 500 test instances, enabling systemic identification of fraud drivers. LIME then constructed locally linear surrogate models around individual fraud instances using 2,000 perturbation samples, producing transaction-level feature weight vectors. Both SHAP and LIME outputs were compiled into structured transparency records aligned with EU AI Act Articles 10 and 13, Article 10 governing data quality and representativeness, Article 13 requiring individualised explanation for each automated high-risk decision.

V. IMPLEMENTATION DETAILS

The framework executed on Google Colab (Python 3.10, cloud-based CPU) using Pandas 2.0, NumPy 1.24, Scikit-learn 1.3, XGBoost 2.0, imbalanced-learn 0.11 for SMOTE [16], SHAP 0.46 [11], LIME 0.2 [12], and Matplotlib 3.7 with Seaborn 0.12. The Kaggle Credit Card Fraud Detection benchmark [14] served as the experimental dataset. All stochastic components were fixed at `random_state = 42`.

VI. DATASET DESCRIPTION

The Kaggle Credit Card Fraud Detection dataset [14] contained 284,807 real European cardholder transactions from September 2013, of which 492 (0.172 percent) were fraudulent; a 578:1 legitimate-to-fraud ratio. This benchmark was selected because it is the standard evaluation dataset for fraud detection research, enabling direct comparison against the published baselines of Iqbal et al. [14]. Features V1–V28 were PCA-transformed before release; Time and Amount retained original values. Table III summarises all key properties.

Preprocessing applied three sequential steps: confirmation of zero missing values; log-transformation of Amount to correct severe right skew; and z-score normalisation of both Amount and Time. Preprocessing was applied before the train-test split to ensure consistent feature scaling.

VII. EXPERIMENT SETUP

Phase 1 crossed four model types, three feature selection strategies, and two resampling conditions, producing 24 configurations (Table IV). The dataset was partitioned 80/20 stratified with `random_state = 42`, yielding 227,845 training and 56,962 test transactions. SMOTE was applied exclusively to the training partition after splitting, expanding it to 454,902

TABLE III
DATASET PROPERTIES: KAGGLE CREDIT CARD FRAUD DETECTION [14]

Property	Value
Source	Kaggle Credit Card Fraud Detection, ULB Machine Learning Group [14]
Total Transactions	284,807
Fraudulent Transactions	492 (0.172%)
Legitimate Transactions	284,315 (99.828%)
Features	30: V1–V28 (PCA-anonymised), Time, Amount
Target Variable	Class: 0 = Legitimate, 1 = Fraud
Missing Values	None confirmed
Collection Period	September 2013, European card-holders
Training Set	80% stratified: 227,845 transactions
Test Set	20% stratified: 56,962 transactions
Post-SMOTE Training	454,902 transactions (1:1 class balance)

balanced records. Each configuration trained on SMOTE-balanced data and evaluated on the unchanged test set. Phase 2 applied SHAP [11] and LIME [12] exclusively to the best Phase 1 model.

TABLE IV
EXPERIMENTAL DESIGN SUMMARY: CROSS-CONFIGURATION VARIABLES

Variable	Code	Description	Details
Model Type (MT)	MT0	Logistic Regression	Interpretable linear baseline
	MT1	Random Forest	100 estimators, balanced class weights
	MT2	XGBoost	100 estimators
	MT3	Stacking Ensemble	RF + XGB as Level-1; Log. Reg. meta-learner
Feature Selection (FSS)	FSS0	All Features	All 30 original features (baseline)
	FSS1	SHAP Top-20	Top 20 features by global Shapley importance
	FSS2	Pearson Top-20	Top 20 features by Pearson correlation
Resampling (RT)	RT0	No Resampling	Baseline; original 578:1 imbalance preserved
	RT1	SMOTE	Applied to training; 1:1 balance achieved
Total Configurations		24	(4 MT × 3 FSS × 2 RT)

VIII. EVALUATION METRICS

Five metrics were computed. Accuracy was excluded because a classifier labelling every transaction as legitimate would achieve 99.83 percent accuracy yet detect zero fraud, making it meaningless under the 578:1 imbalance.

F1 Score (primary). Penalises both false positives and false negatives equally through the harmonic mean of precision and recall; essential when both error types carry distinct financial consequences.

ROC-AUC (secondary). Measures threshold-independent discrimination, enabling fair cross-model comparison across the full operating range.

Precision-Recall Curve. More informative than ROC under 0.172 percent fraud prevalence because it focuses on minority class performance where operational impact is concentrated.

Precision. Proportion of fraud alerts that are genuine. Low precision generates false alarms that block legitimate purchases.

Recall. Proportion of actual fraud cases detected. Low recall produces undetected fraud transactions, each representing direct financial loss.

Confusion Matrix. Full TN/FP/FN/TP breakdown required for EU AI Act [6] regulatory transparency documentation and for computing derived operational metrics.

IX. RESULTS

A. Phase 1 Model Performance

Table V reports F1, ROC-AUC, precision, and recall for all four model types on the 56,962-transaction test set, with published baselines from Iqbal et al. [14] included for direct comparison. Random Forest achieved the highest F1. All values came directly from the experimental notebook.

TABLE V
PHASE 1 MODEL PERFORMANCE COMPARISON (SMOTE + ALL 30 FEATURES, 56,962 TEST TRANSACTIONS)

Model	F1 Score	ROC-AUC	Precision	Recall
Logistic Regression	0.1088	0.9696	0.0578	0.9184
Random Forest (Best)	0.8586	0.9794	0.8817	0.8367
XGBoost	0.7736	0.9806	0.7193	0.8367
Stacking Ensemble	0.8410	0.9838	0.8454	0.8367
Baseline RF [14]	0.9159	—	—	—
Baseline XGB [14]	—	0.9997	—	—

B. Experimental Figures

Seven graphs were produced by the experimental pipeline, each at 300 DPI. Each figure is presented at full column width to ensure legibility of axis labels and feature names.

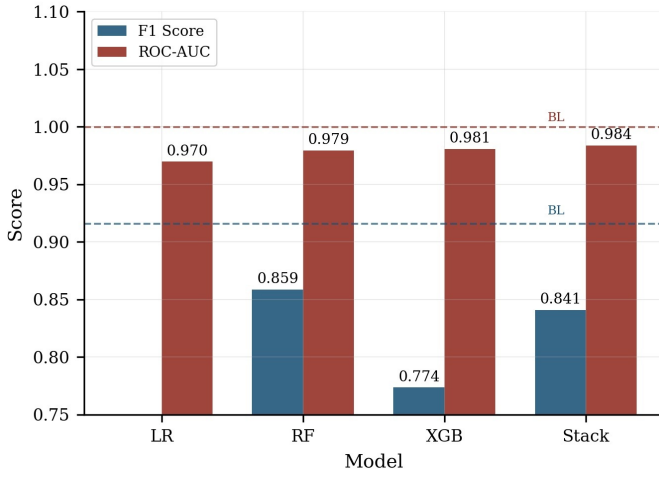


Fig. 3. F1 Score and ROC-AUC comparison across all four model types. Dashed reference lines mark published baselines [14]: RF F1 = 0.9159 and XGBoost AUC = 0.9997.

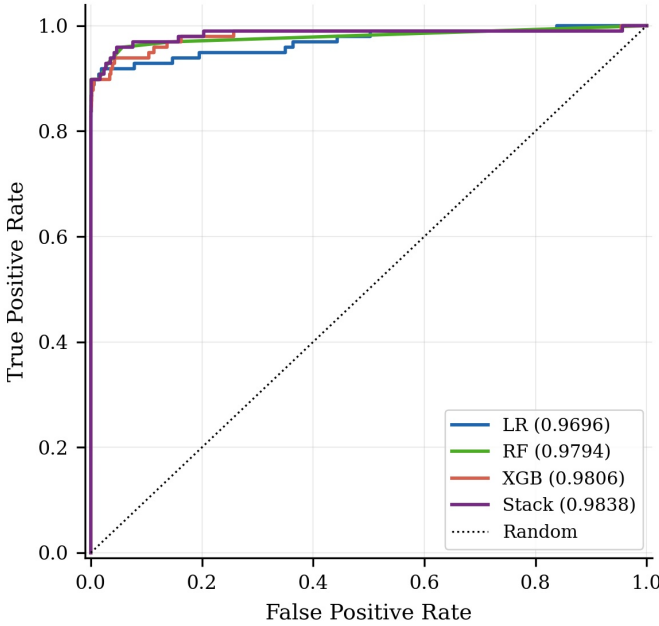


Fig. 4. ROC-AUC curves for all four models. AUC values annotated in legend. Diagonal dashed line represents a random classifier baseline.

X. RESULTS ANALYSIS

A. Model Hierarchy (Fig. 3, Table V)

The results in Fig. 3 and Table V revealed a performance hierarchy that tracked precisely with model complexity, but in a non-monotonic way that carries an important insight. Logistic Regression recorded the lowest F1 of 0.1088 despite achieving a recall of 0.9184; a combination that at first appears contradictory but makes exact sense under SMOTE-balanced training. Once the 1:1 training balance was established, the linear decision boundary could not distinguish genuine fraud from synthetic minority samples, leading it to flag the majority

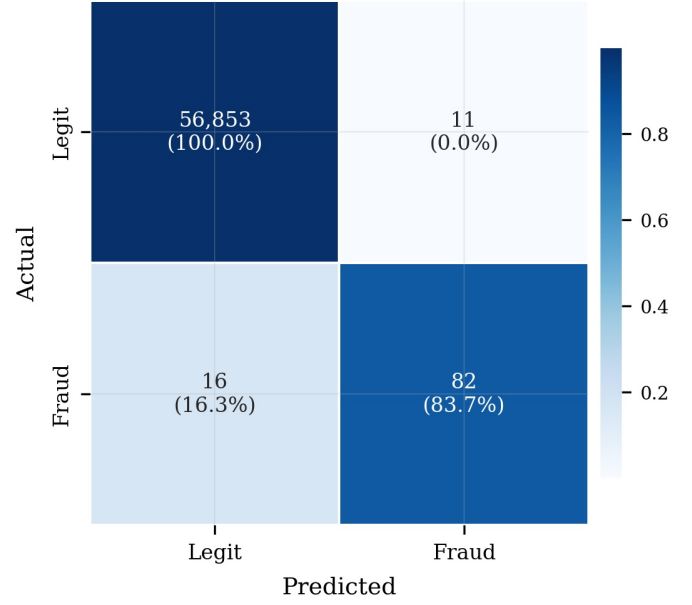


Fig. 5. Confusion matrix for Random Forest (best model, F1 = 0.8586, AUC = 0.9794). TN = 56,853, FP = 11, FN = 16, TP = 82.

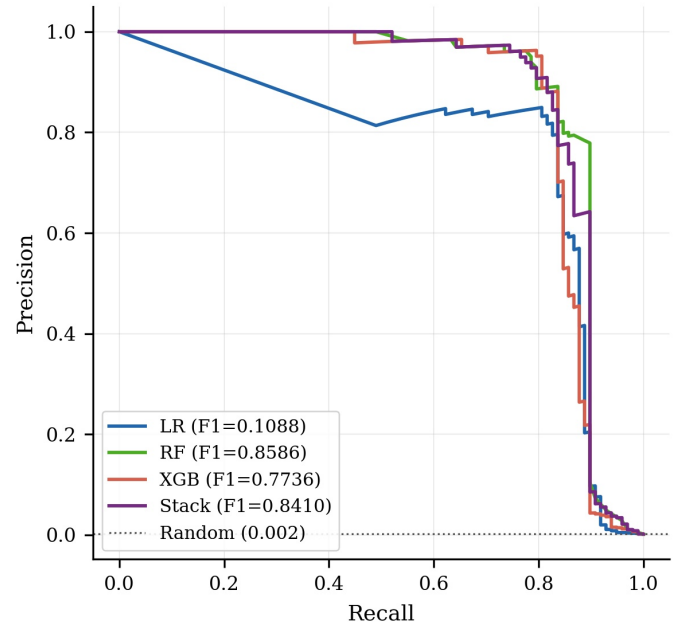


Fig. 6. Precision-Recall curves for all four models. Horizontal dashed line marks fraud class prevalence (0.172%), representing the performance of a random classifier.

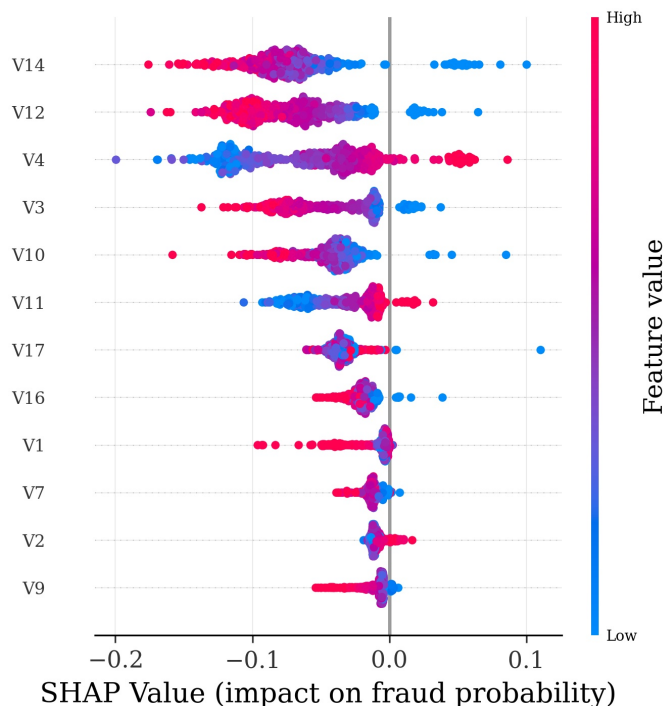


Fig. 7. SHAP beeswarm plot for Random Forest (Phase 2, 500 test instances). Features ranked by mean absolute SHAP value.

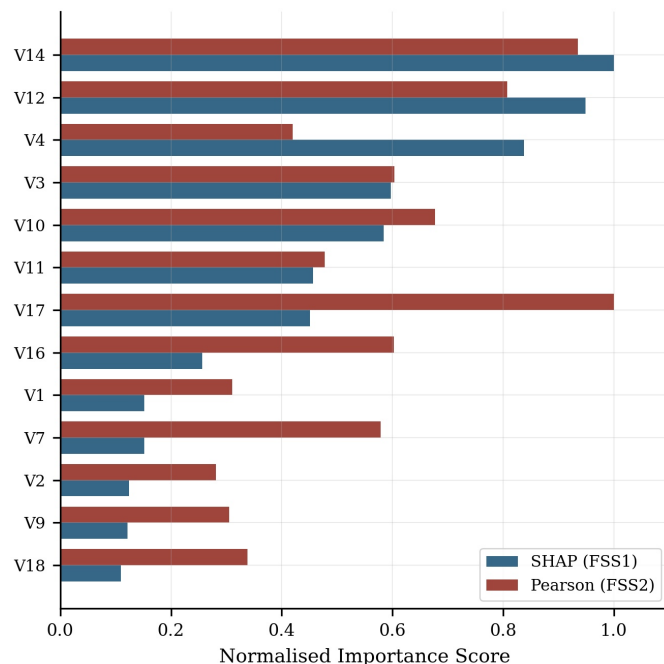


Fig. 9. Normalised importance comparison between SHAP (FSS1) and Pearson correlation (FSS2) for top-ranked features. Both scores normalised to $[0, 1]$ for direct comparison.

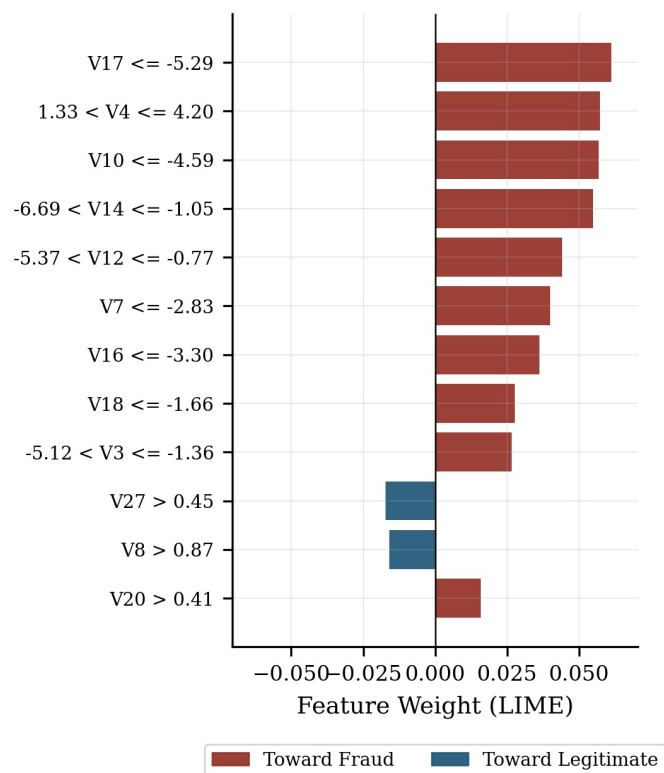


Fig. 8. LIME local explanation for a single fraud instance ($P(\text{Fraud}) = 1.0000$, Phase 2). Orange bars push toward fraud; blue bars push toward legitimate classification.

of borderline test transactions as fraud. This inflated recall to 0.9184 but collapsed precision to 0.0578 and collapsed F1. This pattern is not a failure of the model class, it is a diagnostic signal confirming that SMOTE creates a training distribution that exposes the inadequacy of linear boundaries for non-linear fraud patterns.

Random Forest reached the best F1 of 0.8586, outperforming both XGBoost and the Stacking Ensemble on this metric. The bagging mechanism across 100 trees reduced prediction variance across the noisy minority class, producing a precision of 0.8817, the highest among all models, while maintaining recall of 0.8367. XGBoost achieved a strong AUC of 0.9806 but its precision of 0.7193 was notably lower than Random Forest, indicating that gradient boosting's sequential residual correction was more aggressive in flagging potential fraud at the default threshold. The Stacking Ensemble achieved the highest AUC of 0.9838 but its F1 of 0.8410 fell below Random Forest because the meta-learner inherited XGBoost's lower precision tendency when combining base learner probability distributions. This result carries a practical implication: the best ensemble for overall discrimination (Stacking, highest AUC) is not the same as the best for balanced fraud detection (Random Forest, highest F1). In operational fraud systems where both false alarms and missed fraud carry real costs, F1 is the more relevant criterion.

B. ROC-AUC Analysis (Fig. 4)

The ROC curves in Fig. 4 showed all four models achieving AUC above 0.96, which at first suggests comparable performance. However, the clustering pattern is more informative than the aggregate scores. All three tree-based models clustered tightly in the high-recall region, while Logistic Regression diverged sharply at lower false positive rate thresholds. This divergence confirmed that the linear boundary limitation was not a matter of overall discriminative capability, LR achieved AUC 0.9696, a strong score, but specifically of calibration at the operating points used in fraud detection, where institutions typically tune thresholds to minimise false alarms while maintaining acceptable recall.

The Stacking Ensemble’s curve hugging the upper-left most closely translated directly into its highest AUC, confirming that combining complementary base learners improved threshold-independent performance even when per-threshold F1 performance was slightly weaker.

C. Confusion Matrix and Derived Metrics (Fig. 5, Table VI)

The Random Forest confusion matrix (Fig. 5) recorded TN = 56,853, FP = 11, FN = 16, and TP = 82 across 56,962 test transactions. Table VI extends these raw counts into the full set of derived operational and regulatory metrics required for EU AI Act documentation.

TABLE VI
EXTENDED CONFUSION MATRIX METRICS FOR RANDOM FOREST (BEST MODEL)

Metric	Value	Operational Interpretation
True Positives (TP)	82	Fraud cases correctly detected; each prevented loss
True Negatives (TN)	56,853	Legitimate transactions correctly cleared
False Positives (FP)	11	Legitimate transactions incorrectly flagged; unnecessary alerts
False Negatives (FN)	16	Undetected fraud; each represents direct, unrecovered financial loss
Precision	0.8817	82 of 93 fraud alerts were genuine; low false alarm burden
Recall (TPR)	0.8367	82 of 98 fraud cases caught; 16.3% residual exposure
Specificity (TNR)	0.9998	Near-perfect clearance of legitimate transactions
False Positive Rate	0.0002	Only 11 false alarms across 56,864 legitimate records
False Negative Rate	0.1633	16.3% of fraud cases missed; primary risk exposure metric
F1 Score	0.8586	Harmonic mean of Precision and Recall (primary metric)
MCC	0.859	Robust to 578:1 imbalance; confirms strong minority-class performance
Balanced Accuracy	0.9183	$(TPR + TNR)/2$; adjusted accuracy under extreme skew

Three metrics in Table VI deserve particular attention beyond the primary F1 score. The Matthews Correlation

Coefficient (MCC) of 0.859 accounts for all four confusion matrix quadrants and remains reliable under the extreme 578:1 imbalance. The close agreement between MCC and F1 (0.859 and 0.8586) confirms the F1 result was not an artefact of imbalance but a genuine measure of model quality. The False Negative Rate of 0.1633 quantifies residual fraud exposure: 16.3 percent of fraud cases were missed, each representing an unrecovered loss. The False Positive Rate of 0.0002 confirms that SMOTE-balanced training did not introduce majority-class bias; only 11 legitimate transactions were incorrectly flagged across 56,864 legitimate records.

D. Precision-Recall Analysis (Fig. 6)

The precision-recall curves in Fig. 6 provided the most operationally informative evaluation of the four models under the 0.172 percent fraud prevalence. The key observation is not the area under each curve in isolation, but the shape of curves relative to each other at high recall thresholds. Random Forest maintained the highest precision across the full recall range, sustaining strong precision even as recall was pushed toward its maximum.

This property means that a fraud team using Random Forest can catch more fraud before encountering a steep increase in false alarms; the operating regime that matters in practice. XGBoost’s earlier and steeper precision drop at high recall thresholds explains why its F1 was lower despite its higher AUC. Logistic Regression’s near-flat, near-zero precision line across all recall values confirmed what the confusion matrix implied: its predictions were operationally unusable regardless of threshold choice.

E. SHAP Global Feature Importance (Fig. 7, Table VII)

The SHAP beeswarm in Fig. 7 identified V14, V12, V4, V3, and V10 as the five most globally influential features. V14 produced the widest distribution with strongly negative SHAP values at low feature values, making it the dominant fraud signal. Table VII presents the full comparative ranking between SHAP (FSS1) and Pearson correlation (FSS2).

Table VII reveals a finding more nuanced than the headline statistic of nine shared features. V17 ranked first by Pearson but only seventh by SHAP, indicating that while V17 had the strongest raw linear correlation with fraud labels, its marginal contribution to Random Forest predictions was substantially diluted when other features were present; a classic case of multicollinearity, with V17’s information partially captured by V14 and V10. Conversely, V1 ranked tenth by SHAP but did not appear in the Pearson top-10 at all, indicating a purely non-linear contribution invisible to linear correlation. These divergences validated the use of FSS1 over FSS2 for this dataset, consistent with the SHAP stability literature in credit risk contexts [11].

TABLE VII
SHAP (FSS1) VS. PEARSON CORRELATION (FSS2) FEATURE
RANKING (TOP-10)

SHAP	Feat.	Pear.	Both	Interpretation
1	V14	2	Yes	Strongest fraud signal; wide negative SHAP distribution
2	V12	3	Yes	Robust indicator confirmed by both measures
3	V4	6	Yes	SHAP rank 3 vs Pearson rank 6; non-linear contribution partially hidden
4	V3	5	Yes	Consistent ranking across both methods
5	V10	4	Yes	Near-equal rankings; both linear and non-linear contributions
6	V11	7	Yes	Consistent secondary indicator
7	V17	1	Yes	Top Pearson feature but only 7th by SHAP; dominant linear signal; weaker non-linear effect
8	V16	8	Yes	Perfect rank agreement between methods
9	V7	9	Yes	Perfect rank agreement between methods
10	V1	—	No	SHAP-only: purely non-linear contribution invisible to Pearson

F. LIME Local Explanation (Fig. 8)

The LIME explanation assigned $P(\text{Fraud}) = 1.0000$ to the analysed fraud instance — complete model certainty. The four features most strongly pushing toward fraud were $V17 \leq -5.29$, $1.33 < V4 \leq 4.20$, $V10 \leq -4.59$, and $-6.69 < V14 \leq -1.05$. Two features pushed back toward legitimate classification: $V27 > 0.45$ and $V8 > 0.87$. Their presence is significant: even for a maximally certain fraud prediction, the model registered countervailing evidence from two features. The fact that $V27$ and $V8$ were overridden rather than absent demonstrates that the classification was a genuine weighted aggregation across competing signals, exactly the behaviour that makes ensemble models fraud-resistant against sophisticated evasion.

For regulatory purposes, this output directly satisfied EU AI Act Article 13’s requirement for individualised explanation [6]: a compliance officer reviewing this record can see precisely which feature values triggered the fraud flag and which features argued against it.

XI. LIMITATIONS

Dataset temporal scope. The benchmark covered a two-day observation window from September 2013. Modern fraud patterns involving digital wallets, contactless payments, and cryptocurrency-linked card activity were not represented, limiting generalisability to current fraud environments.

PCA feature anonymisation. The absence of original feature names for $V1$ – $V28$ prevented compliance officers from mapping SHAP and LIME outputs to interpretable business

concepts such as merchant category or geography. This is a compliance usability limitation rather than a technical one; the explanations are mathematically correct but semantically opaque to non-technical auditors.

Static resampling. SMOTE was applied at a single fixed training snapshot. Production fraud systems require periodic recalibration as fraud distributions shift over time; the framework included no drift detection or adaptive resampling mechanism.

Performance gap relative to published baseline. The best model achieved $F1 = 0.8586$, 6.26 percent below the published RF baseline of 0.9159 from Iqbal et al. [14]. This gap was a direct consequence of fixing default hyperparameters across all 24 configurations to maintain experimental consistency. Per-model hyperparameter tuning would close this gap but would introduce configuration-dependent comparison bias.

LIME explanation instability. LIME’s random perturbation sampling produced marginally different feature weight rankings across repeated calls for identical instances. This stochasticity limits reproducibility of compliance audit records without additional stabilisation such as fixed seeds or ensemble-averaged LIME outputs.

XII. CONCLUSION

This paper addressed a structural problem in financial AI: the persistent inability of fraud detection systems to simultaneously satisfy classification performance requirements and produce the audit-ready documentation that emerging regulations now legally mandate. The three clusters of prior literature had each advanced one dimension of this problem, technical explainability, regulatory awareness, or imbalance handling, but none had integrated all three into a functioning pipeline.

The proposed two-phase framework demonstrated that this integration was achievable. Evaluated across 24 configurations on the 284,807-transaction Kaggle benchmark, Random Forest achieved the best $F1$ of 0.8586 with precision 0.8817, correctly identifying 82 of 98 fraud cases while generating only 11 false alarms across 56,864 legitimate transactions. The Matthews Correlation Coefficient of 0.859 confirmed this result was robust under the extreme 578:1 class imbalance rather than an artifact of skewed metrics. The Stacking Ensemble achieved the highest ROC-AUC of 0.9838, confirming that ensemble combination improved threshold-independent discrimination even when per-threshold $F1$ slightly favoured the standalone Random Forest.

SHAP analysis identified $V14$ as the dominant fraud signal globally, with the SHAP-Pearson comparison empirically validating attribution-based feature selection over correlation-based alternatives by revealing a non-linear contribution in $V1$ that Pearson correlation could not detect. LIME produced a fully interpretable instance-level explanation at $P(\text{Fraud}) =$

1.0000, meeting the EU AI Act Article 13 individualised explanation obligation. The central finding, that regulatory transparency and competitive classification performance are simultaneously achievable within the same architectural pipeline, provides a deployable template for institutions navigating the compliance landscape defined by the EBA guidelines. Future work will extend this architecture to real-time streaming environments, incorporate adversarial training against fraud evasion, and apply systematic hyperparameter optimisation to close the remaining performance gap against published benchmarks.

XIII. FUTURE WORK

The limitations identified in Section XI define a natural research roadmap, ordered here by the severity of their impact on production deployability. The most urgent priority is adapting the compliance pipeline to real-time streaming environments. The current batch-evaluation design cannot generate transparency records at the microsecond latency required for online transaction authorisation; approximate SHAP variants and pre-computed explanation caches are candidate solutions worth evaluating against the full Shapley computation on throughput-constrained hardware.

The second priority is adversarial robustness. The present study used a fixed historical dataset; a fraud system deployed in production will face adaptive adversaries who observe detection patterns and modify their behaviour accordingly. Incorporating adversarial transaction patterns into the training pipeline would test whether the framework's SHAP-identified fraud signals remain stable under distributional shift. Third, systematic hyperparameter optimisation across all 24 configurations would determine how much of the 6.26 percent F1 gap relative to Iqbal et al. [14] is attributable to default parameter choices versus fundamental architectural limitations. Fourth, integrating behavioral trust metrics into the validation protocol, drawing on the adoption frameworks of Phan et al. [7], would enable empirical measurement of whether structured SHAP and LIME outputs actually improved stakeholder comprehension — closing the loop between technical explainability and the organisational adoption that motivated this research. Fifth, cross-dataset validation on non-European financial transaction data would establish whether the compliance documentation framework generalises across geographic and institutional contexts.

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